

CO₂ emissions and financial development in an emerging economy: An augmented VAR approach



Faiza Abbasi*, Khalid Riaz

Department of Management Sciences, COMSATS Institute of Information Technology, Park Road, Islamabad, Pakistan

HIGHLIGHTS

- In economies where structural transformation of financial sector is in early stages.
- Financial development does not aid in mitigating CO₂ emissions rather it increases it.
- CO₂ emissions rise as per capita income rises.
- Government should devise comprehensive and realistic mitigation strategies.

ARTICLE INFO

Article history:

Received 5 August 2015

Received in revised form

30 November 2015

Accepted 14 December 2015

Available online 22 December 2015

Keywords:

CO₂ emissions

Financial development

Pakistan

ARDL

Augmented VAR

VECM

ABSTRACT

This paper explores the influence of economic and financial development on carbon emissions in a small emerging economy. The study employs ARDL approach to investigate the long run relationship between carbon emissions and a set of economic and financial variables, an Error Correction Model (ECM) to capture the short run dynamics, Granger causality in an augmented VAR framework to check the causality direction, and variance decomposition based on an estimated Vector Error Correction Model (VECM) to determine the relative contributions of economic and financial variables to the evolution of per capita carbon emissions. The periods considered were the full sample (1971–2011), and a reduced sample sub-period (1988–2011) that corresponded to greater liberalization and financial sector development. The financial variables played a role in emission mitigation only in the latter period where greater degree of liberalization and financial sector development occurred. Even then the relative magnitude of emissions mitigation attributable to financial variables was much smaller compared to the emissions raising impact of rising per capita incomes. This underscores the need for adopting other mitigation policies for reducing carbon footprints in those emerging economies where a sufficient degree of financial deepening and financial sector development has not yet taken place.

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1. Introduction

There is an extensive literature on the relationship between economic development and environmental degradation i.e. the Environmental Kuznets Curve (EKC) hypothesis. According to this hypothesis, there exists a positive relationship between economic development and environmental degradation initially as countries attempt to raise standards of living from initial low levels, even if that involves some environmental degradation. But as the economies attain higher levels of development, achieving better environmental outcomes becomes increasingly important. Thus the EKC has an inverted U shape. Results from several studies support

the existence of an inverted U-shaped relationship between economic development and environmental degradation. For example, [Shafik \(1994\)](#) provides evidence of an inverted EKC for suspended particulate matter (SPM), deforestation rate, and SO₂; see [Grossman and Krueger \(1991\)](#) for SO₂, SPM, and dark matter; and [Orubu and Omotor \(2011\)](#) for SPM.

In recent years, the climate change and its implications for human race have emerged as important themes in global environmental debate. This has focused greater attention on the underlying causes of climate change that include increased concentration of greenhouse gases (GHGs) due to anthropogenic activity. An important GHG is Carbon dioxide (CO₂) that accounts for 75% of GHG emissions (WDI, 2010). There is a growing body of literature on the relationship between CO₂ emissions and economic development. For example, several studies have noted the existence of a long-run positive relationship between economic growth, energy use and CO₂ emissions [see [Ang \(2007\)](#) and [Ozturk and Acaravci \(2010\)](#)].

* Corresponding author.

E-mail addresses: faiza.abbasi@live.com (F. Abbasi), kriaz100@gmail.com (K. Riaz).

It may be overly simplistic to study the relationship between emissions and growth in isolation to other aspects of economic development. The latter is a complex process that causes structural changes in the economy. An important change that accompanies growth is in the size and structure of the financial sector that provides investments that are essential for growth but also for adopting energy conserving, green technologies. These structural changes in the financial sector have been documented in literature. For example, as economies develop, their financial sectors represent larger shares of the overall economy. This is true not only of the banking sector but also of the stock market. Moreover, the stock market becomes more important relative to the banking sector (Demircuc-Kunt et al., 2012). However, it is not clear how this structural transformation affects the evolution of CO₂ emissions.

One school of thought suggests that CO₂ emissions increase because financial development alleviates the credit constraints and helps the economic output to expand which results in more energy consumption and higher CO₂ emissions. According to Sadorsky (2010), efficient financial intermediation encourages the customers to take loans and buy big ticket items like automobiles etc., which increases CO₂ emissions. Dasgupta et al. (2001) suggest that development of stock market results in low financing cost, easing the liquidity limitations for listed enterprises, allowing them to expand output, increase energy consumption and hence CO₂ emissions.

However, the other school of thought maintains that financial sector development boosts investment in technologies that are energy efficient and thus reduces CO₂ emissions. Frankel and Romer (1999) argued that financial development in a country draws more foreign direct investment (FDI) and causes higher level of Research and Development (R&D) that leads to better environment. Talukdar and Meisner (2001), Meilnik and Goldemberg (2002), Wang and Jin (2007) and Bello and Abimbola (2010) found that increase in FDI brought technological improvement that led to lower emissions. Tamazian et al. (2009) found that financial development helped the listed companies to increase energy efficiency by adopting new technologies.

The recent literature is also divided on the impact of financial development on CO₂ emissions. Almost all the studies have taken FDI and/or domestic credit to private sector as the proxies to measure financial development. The literature shows mixed results where majority of the studies support the reducing impact of private sector credit and the increasing impact of FDI on CO₂ emissions. Ren et al. (2014) studied the CO₂ emissions in various industrial sectors of China. They found that FDI increased emissions in the industrial sectors because of lack of efficiency in resource utilization and technical level. Lau et al. (2014) investigated the presence of EKC for Malaysia in presence of FDI and found that in the long run, FDI increased CO₂ emissions.

Shahbaz et al. (2013a, 2013b, 2013c) tested the EKC hypothesis for South Africa and studied the impact of economic and financial variables on CO₂ emissions. The results revealed that domestic credit to private sector decreased emissions in both long and short run for South Africa. Shahbaz et al. (2013a, 2013b, 2013c) studied the linkages between CO₂ emissions, economic and financial variables in Indonesia. The study found that real domestic credit to private sector reduced CO₂ emissions in the long run. Moreover, Shahbaz et al. (2013a, 2013b, 2013c) explored financial development impact on CO₂ emissions for Malaysia. The empirical results suggested that private sector credit had reducing impact on CO₂ emissions whereas FDI had increasing impact on emissions.

Some recent empirical studies support the positive or no-energy reducing impact of private sector credit on emissions. Khan et al. (2014) studied the link between CO₂ emissions and some economic and financial variables (FDI, Money and quasi money (M2), Liquid liabilities (M3), domestic credit provided by banking sector and private sector credit) for South Asia. The results

suggested the positive impact of all the financial variables on CO₂ emissions in the long run, with no impact in short run. This supports the argument that foreign investment and provision of liquidity through banking sector facilitates more investment in businesses/output expansion leading towards more consumption of energy and hence more emissions.

Zhang (2011) made an important contribution to this debate by separately considering the scale and efficiency effects of financial development on CO₂ emissions for China. We extend Zhang (2011) in two ways. First, we investigate whether and how the relationship between CO₂ emissions and financial development changes for a small emerging economy. Pakistan is taken as a case in point. Second, Zhang (2011) relies on Johansen cointegration approach that assumes all variables are integrated of the same order. We use the ARDL approach instead that allows variables to have different orders of integration. Since economic series may exhibit mixed patterns of integration, our approach has a wider applicability.

2. Financial development in Pakistan

Pakistan belongs to the lower middle income group, as per WDI, 2015 (see Fig. 1). Like many other developing countries, Pakistan introduced financial sector reforms under the structural adjustment program of IMF and World Bank during the late 1980s. The reforms were aimed at instilling competition, improving efficiency and productivity of the financial markets. The results of this liberalization were impressive, accompanied by an increased number of privately owned banks, investment banks, micro-financing institutions and mutual funds. Pakistan's financial sector is still in the earlier stages of development, the banking sector has undergone remarkable growth. Although the efficiency in provision of loans to different sectors especially to private sector has not shown any momentum, it has been unstable for years (see Fig. 10).

After the financial liberalization, the growth of stock market was remarkable but the lack of stability in the market and in the country was evident. Karachi Stock Exchange (KSE) was the first stock exchange of the country, it started operating in 1947 and undergone steady progress as more companies got listed in KSE until 1970s (see Fig. 11). The political instability in the country and the nationalisation policies of the government during 1970s, made it lost its momentum. It gained momentum again in 1974 when more and more companies started listing in the stock exchange but the political uncertainty and the consequential martial law badly affected the investment atmosphere. In early 1990s, stock market recovered as it was opened to foreign investors but it has not performed consistently due to a number of factors including political instability, terrorism, inflation, unemployment and budget deficits etc. The lack of maturity and the volatility of the financial sector affected the overall business environment, investment, growth and CO₂ emissions.

The liberalization policies after mid 1980s resulted in a moderate increase of FDI in Pakistan (see Fig. 7). However, it lost its momentum after mid 1990s when the US assistance to Pakistan slowed down significantly. The actual decline of FDI as a percentage of GDP occurred between 1995 and 2000 when sanctions were imposed by US after Pakistan's nuclear tests in 1998. FDI share in GDP rose very sharply after 9/11 period when Pakistan joined the war against terrorism, but it has shown considerable volatility over the years, reflecting Pakistan's relationship with the major world powers and uncertain political and economic condition of the country.

The financial sector is less developed as compared to some other developing economies. Figs. 1–6 provide graphs of the major macroeconomic and financial indicators averages for different income groups (Low, middle, high) in comparison to Pakistan

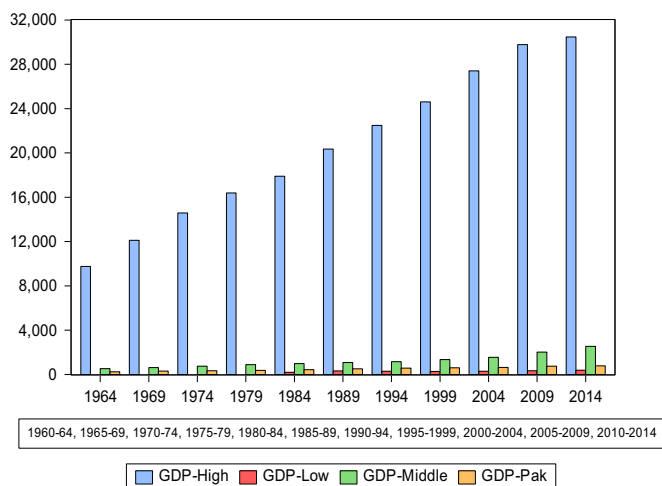


Fig. 1. 5-yearly period averages comparison_High, Middle, Low Income Economies_GDP Per Capita.

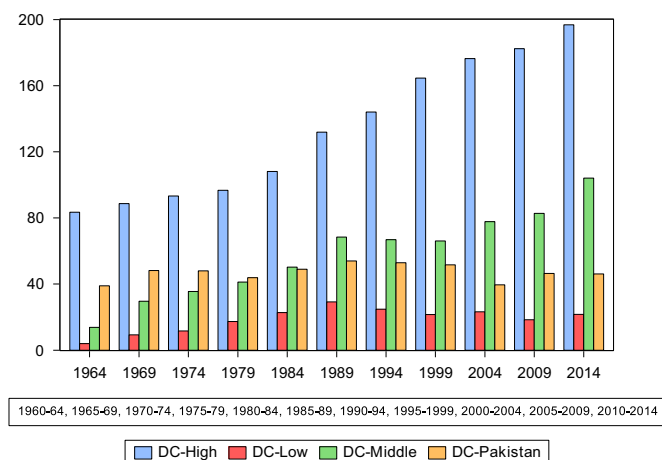


Fig. 2. 5-yearly period averages comparison_ Total Credit (% of GDP).

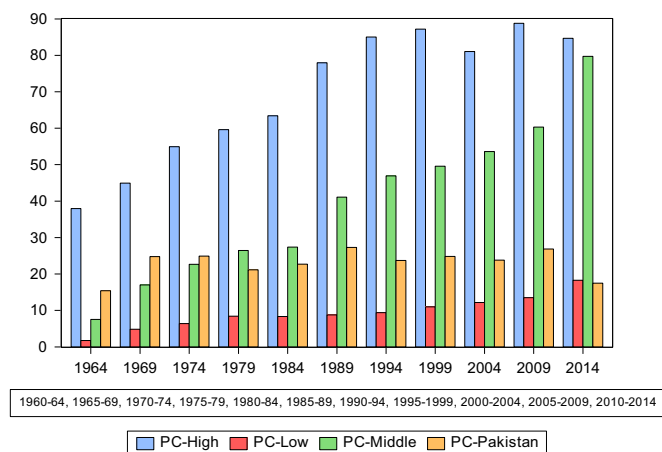


Fig. 3. 5-yearly period averages comparison_ Private Credit (% of GDP).

(5-yearly period averages starting from 1960–64 to 2010–14). These comparisons help consider the financial development in Pakistan in relation to other developing economies.

The banking sector development indicators (Domestic credit (% of GDP) and private sector credit (% of GDP)) highlight the contrast between Pakistan's degree of financial development with that of the other middle income countries. Overall, the developing country credit indicators (Figs. 2 and 3) showed a clear upward

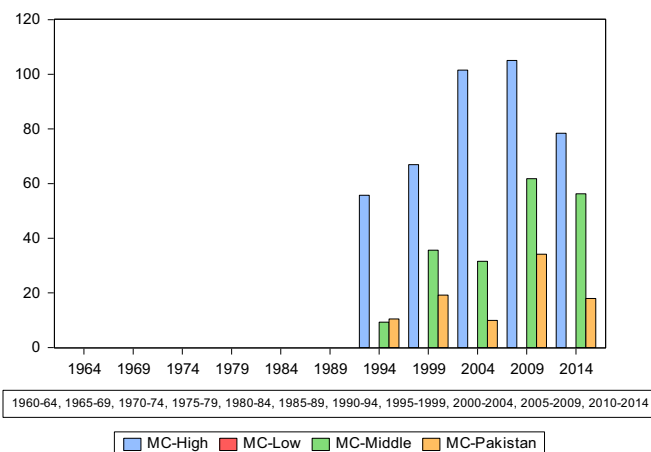


Fig. 4. 5-yearly period averages comparison_Market Capitalization (% of GDP).

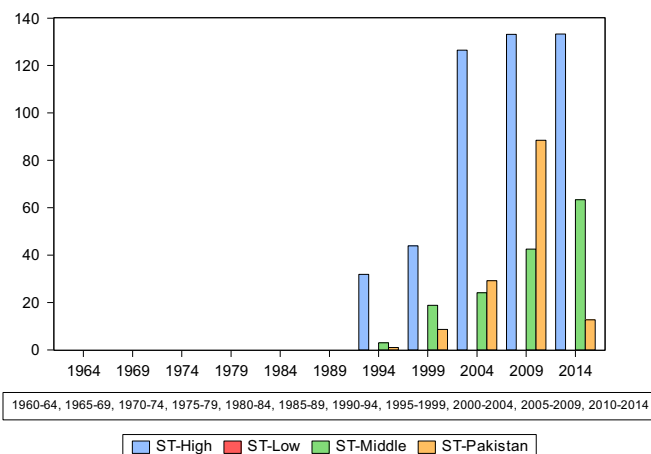


Fig. 5. 5-yearly period averages comparison_Stock Market Turnover (% of GDP).

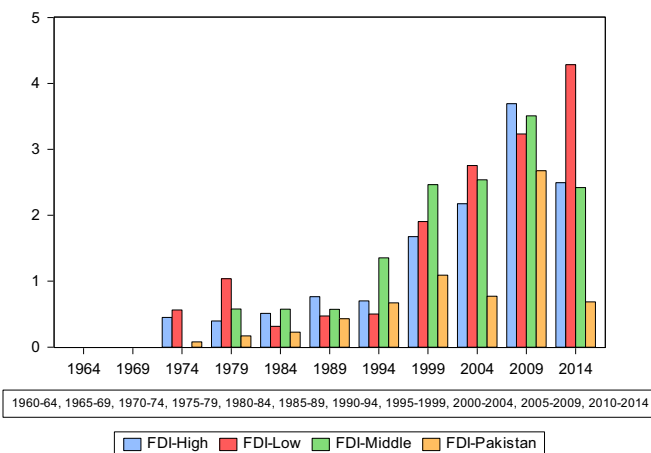


Fig. 6. 5-yearly period averages comparison_ FDI (% of GDP).

trend over the decades whereas for Pakistan, there was a steady pattern showing no growth. The availability of total credit as well as the availability of credit to private sector in Pakistan was significantly lower compared to middle income countries average. For private sector credit, there was a clear divergence with the corresponding indicator for middle income developing countries. Specifically, Pakistan's private credit share in GDP decline quite in the last five years while the average of middle income countries

increased very sharply over the same period.

Like the High and Middle income countries, the stock market development indicators—(market capitalization (% of GDP) and stocks turnover/traded (% of GDP) – in Pakistan also) exhibited an increasing trend. Interestingly, stock market in Pakistan did perform considerably well in the last decade even surpassing the other middle income economies. During the second half of the previous decade, stock market turnover exceeded the average of middle income countries. But it dropped to below that average following the 2008 crisis (see Fig. 5). The market capitalization in Pakistan remained below middle income country average during the study period.

A well-developed financial sector is considered as one of the underlying causes to attract FDI in a country. FDI has also seen growth for all the income groups especially for Pakistan during the period 2005–2009. However, the overall level of FDI was considerable lower in Pakistan compared to even the middle income countries. This difference has become more pronounced in the last five years (see Fig. 6).

In summary, there were significant differences in level of economic and financial development of Pakistan as compared to the middle income group as a whole. Pakistan's financial development indicators and FDI inflows had seen rapid improvement only in the last decade. The earlier studies that investigated the relationship between emissions and financial sector development have largely focused on economies (e.g. China, Malaysia, and South Africa etc.) that had different (even higher) level of economic and financial development. It is not clear whether the results of those studies carry over in a straight forward manner to the smaller emerging economies where the level of financial development may be considerably lower. Our study attempted to fill this important gap by studying the relationship between carbon emissions and financial development for a small emerging economy, Pakistan, which had a relatively lower level of financial development. The results of our investigation would hopefully inform the larger debate on the nature of relationship between emissions and development of the financial sector.

The role of economic and financial variables in the evolution of CO₂ emissions has received less attention in research literature on Pakistan. But in recent years, there has been greater interest in the topic [for example see Alam et al. (2007), Shahbaz et al. (2011), Shahbaz (2011), Islam et al. (2011)]. The studies on Pakistan majorly focused on the impact of economic development on CO₂ emissions. Only Shahbaz et al. (2011) investigated the impact of financial development on CO₂ emissions for Pakistan. He used market capitalization as a proxy for financial development. The present study attempts to contribute to this literature by considering the full range of financial development indicators – incorporating both the scale and efficiency effects in the spirit of Zhang (2011) – and their impacts on CO₂ emissions.

Pakistan is the 31st highly CO₂ emitting country. Its emissions have increased 8 times more than it was emitting in 1971 (US EIA, 2010).¹ According to a recent list issued by the World Health Organization (2015), Karachi is ranked at number 5 whilst Peshawar and Rawalpindi are ranked at 6 and 7 respectively, among the top 20 most polluted cities in the world. Because of rapid increase in population, urbanization, rising in per capita income, increased industrial activity the energy demand and CO₂ emissions have increased rapidly. In January 2005, Pakistan signed Kyoto Protocol and vowed to enforce Clean Development Mechanism (CDM) for GHGs mitigation. Government of Pakistan also launched 'National Environmental Policy' (NEP) in 2005

(Ministry of Climate Change, Government of Pakistan)² which is aimed at protecting the environment, and improving the quality of life through sustainable development.

This study divides the sample period into two parts where the latter period is characterized with more economic and financial development that helps to interpret results more comprehensively. Moreover, a more robust methodology is used where ARDL approach explores the cointegration relationship between variables irrespective of their order of integration and the augmented Vector Autoregressive (VAR) framework that performs VAR in levels. It preserves the long run information, which would be lost if variables were made stationary by differencing prior to estimation of VAR model.

The rest of the study is organized as follows. Section 3 describes data and empirical methodology. Results and discussion are reported in Section 4 and finally Section 5 concludes the study along with some policy implications.

3. Data and methodology

3.1. Data

The definitions of all the variables in this paper are presented in Table 1. The annual data on CO₂ emissions, GDP, FDI, total credit, private sector credit and stocks traded/turnover was taken from World Development Indicators (WDI, 2015). Market capitalization data was taken from State Bank of Pakistan database.

The per capita CO₂ emissions were hypothesized to be related to the level of economic development (proxied by per capita GDP), financial intermediation development, stock market development, and FDI. Following Zhang (2011), we considered both the scale and the efficiency of financial intermediation development and stock market development. The scale of financial intermediation development was represented by percentage of total credit in GDP while the percentage of private sector credit in GDP was taken as proxy for financial intermediation efficiency. Similarly, the scale effect of stock market development was represented by market capitalization percentage in GDP, and the corresponding efficiency effect was taken to be the stock market turnover percentage in GDP.

Full sample data set is from 1970 to 2011, including all the variables except stock market turnover (% of GDP) because of its shorter time period. Reduced sample data set is from 1988 to 2011 including all the variables. This classification enabled the inclusion of all the variables for testing and also helped to interpret the results more comprehensively as in the latter years, the level of economic and financial development had improved for Pakistan which could better explain the relation of these variables with CO₂ emissions.

3.2. Empirical methodology

This study aimed to investigate the impact of economic and financial development on CO₂ emissions in Pakistan during the period 1970–2011. The study first explored the long run equilibrium relationship between economic, financial and environmental variables using ARDL bounds testing approach. In our view, the ARDL generalizes Zhang (2011) who used Johansen and Juselius (1990) method for testing for co-integration that requires variables to be integrated of the same order [Enders (2010)]. Moreover, the approach used by Johansen and Juselius (1990) is not suitable for studying cointegration when the time series involved are small [Mah (2000)]. Kremers et al. (1992) maintain that in small samples, cointegration cannot be established if series are *I*

¹ US Energy Information Administration.

² www.mocc.gov.pk

Table 1
Variables and data definitions.

Variables	Description/Variables	Definition	Sample Period
Carbon emissions (CO ₂)	Measured in Metric tons per capita	Carbon dioxide emissions are those stemming from the burning of fossil fuels and the manufacturing of cement. They include CO ₂ produced during consumption of solid, liquid, and gas fuels and gas flaring.	1970–2011
Economic development	Proxied by GDP per capita at constant 2000 US dollars	GDP per capita is the Gross domestic product divided by midyear population.	1970–2011
Financial intermediation development	Scale Effect Total Credit (% of GDP)	It includes all credit to various sectors on a gross basis.	1970–2011
	Efficiency Effect Private Sector Credit (% of GDP)	Domestic credit to private sector refers to financial resources provided to the private sector.	1970–2011
Stock market development	Scale Effect Stock market capitalization (% of GDP)	Market capitalization (also known as market value) is the share price times the number of shares outstanding.	1970–2011
	Efficiency Effect Stocks traded/turnover (% of GDP)	It refers to the total value of shares traded during the period.	1988–2011
Foreign direct Investment (FDI)	FDI (% of GDP)	FDI are the net inflows of investment to acquire a lasting management interest in an enterprise operating in an economy other than that of the investor.	1970–2011

(1). The problem is not resolved by using quarterly data over the same time period to increase the number of observations; the length of the study period has to be appropriately increased to improve the robustness of cointegration analysis [Hakkio and Rush (1991)]. This may prove to be a serious limitation for some emerging economies where data limitations are more severe.

By contrast ARDL can be applied to situations where the variables have different orders of integration, and where the time series are short. Both considerations are important if we want to test Zhang (2011) model for other emerging economies where the financial sector may be at a different stage of financial development as compared to China, and the data availability constraints may be more severe resulting in shorter time series of available data.

The ARDL methodology for testing cointegration is superior to earlier approaches such as Engle and Granger (1987), Johansen (1988) and Johansen and Juselius (1990). It can be applied when the series in question are stationary, integrated $I(1)$ or fractionally co-integrated. The earlier approaches, on other hand required all series to be integrated of the same order. Moreover, ARDL gives robust results even in small samples (Pesaran and Shin, 1999), which is a major advantage when the available time series are small, as they typically are in many emerging economies.

The ARDL model for Y_t (per capita CO₂ emissions) in terms of and X_{it} (six other variables discussed above, i.e. GDP per capita and various financial indicators) can be specified as follows:

$$\Delta Y_t = \alpha_0 + \sum_{j=1}^k \alpha_{1j} \Delta Y_{t-j} + \sum_{j=1}^k \alpha_{2j} \Delta X_{1t-j} + \sum_{j=1}^k \alpha_{3j} \Delta X_{2t-j} + \sum_{j=1}^k \alpha_{4j} \Delta X_{3t-j} + \sum_{j=1}^k \alpha_{5j} \Delta X_{4t-j} + \sum_{j=1}^k \alpha_{6j} \Delta X_{5t-j} + \sum_{j=1}^k \alpha_{7j} \Delta X_{6t-j} + \beta_0 Y_{t-1} + \beta_1 X_{1t-1} + \beta_2 X_{2t-1} + \beta_3 X_{3t-1} + \beta_4 X_{4t-1} + \beta_5 X_{5t-1} + \beta_6 X_{6t-1} + \mu_t \quad (1)$$

The subscript 'j' denotes the lag length, Δ is the difference operator and μ_t is the residual term, which is assumed to be normally distributed. The coefficients α_i ($i = 1, \dots, 7$), represent the short run effects, whereas β_j ($j = 0, 1, \dots, 6$) represent the long run effects. The ARDL cointegration tests is based on testing the null hypothesis that all β_i (for $i = 1, \dots, 6$) are zero, against the alternative hypothesis that at least some β_k is non-zero.

Pesaran et al. (2001) has provided the upper and the lower critical bounds, respectively, corresponding to the assumptions that the variables are $I(0)$ (UCB) and $I(1)$ (LCB), against which the calculated value of the F-statistic is compared. A calculated F statistic higher than UCB is taken to be evidence warranting the rejection of the null hypothesis of no cointegration regardless of the order of integration. If a lower value compared to LCB is obtained, the null hypothesis is not rejected. The test is considered

inconclusive between the two critical values. The critical values mentioned above are suitable for larger samples of 1000 observations or more (Pesaran et al., 2001). For smaller samples, between 30 and 80 observations, the set of critical values supplied by Narayan (2004) are used. In the present study, Schwartz criterion was used for determining the optimal lag lengths.

After the evidence of cointegration between variables, the long run relationship is estimated for both samples by the following equation:

$$Y_t = \beta_0 + \sum_{j=0}^k \beta_{1j} X_{1t-j} + \sum_{j=0}^k \beta_{2j} X_{2t-j} + \sum_{j=0}^k \beta_{3j} X_{3t-j} + \sum_{j=0}^k \beta_{4j} X_{4t-j} + \sum_{j=0}^k \beta_{5j} X_{5t-j} + \sum_{j=0}^k \beta_{6j} X_{6t-j} + \mu_t \quad (2)$$

Where β_i ($i = 0, 1, \dots, 6$) are the long run coefficients, μ_t is the error term.

An error correction model (ECM) can be then estimated to highlight the short run dynamics. According to Engle and Granger (1987), the cointegrated variables have an error correction representation. The ECM is implemented by estimating the following equation.

$$\Delta Y_t = \delta_1 + \sum \delta_{2j} \Delta X_{1t-j} + \sum \delta_{3j} \Delta X_{2t-j} + \sum \delta_{4j} \Delta X_{3t-j} + \sum \delta_{5j} \Delta X_{4t-j} + \sum \delta_{6j} \Delta X_{5t-j} + \theta ECT_{t-1} + e_t \quad (3)$$

where the ΔY_t and ΔX_{it} respectively, represent first differences of variables Y_t and various X_{it} , defined above. The ECT_{t-1} is the error correction term (lagged value of residual μ_t from the long run model). The error correction term (ECT_{t-1}) indicates the speed of adjustment of the dependent variable towards its long run equilibrium. It should have a statistically significant coefficient with a negative sign.

For Granger causality tests, the augmented VAR framework due to Toda and Yamamoto (1995) was used. In time series econometrics, Granger causality can be conducted within VAR framework or Vector Error Correction Model (VECM). Sims et al. (1990) demonstrated that for stationary variables VAR model may be specified in levels; if, however, units roots are present but the variables are not cointegrated then VAR in differences is appropriate. For the case where unit roots are present and the variables are also cointegrated, and error correction term is incorporated and VECM is estimated. The potential danger inherent in this approach involving pre-testing for unit roots and cointegration is

that conventional unit root tests have low power to reject the null hypothesis. This can lead to poor inference and possibly poor model selection for causality testing.

The augmented VAR approach by Toda and Yamamoto (1995) provides a way of dealing with this difficulty through modified model specification. Their methodology is robust to the order of integration as well as to the nature of cointegration among variables. In the presence of unit roots, the Wald test statistic does not follow the asymptotic Chi-square distribution, even when the variables are not cointegrated. Differencing may achieve stationarity but would result in loss of information. Under these circumstances, Toda and Yamamoto proposed that the VAR model should be specified in level but its order should be augmented the optimal lag length (k) by the maximal order of integration ($d_{\max}$), yielding a VAR system of order $k + d_{\max}$.

This augmented VAR system is estimated and Granger causality is tested employing the standard Wald Test subject to the condition that the coefficients representing the extra $d_{\max}$ lags are not included. If the null hypothesis is rejected, this is considered to be the rejection of non-causality.

For this study, the Granger causality test based on augmented VAR with $(k + d_{\max})$ lag length was implemented by specifying the following VAR model.

$$\begin{pmatrix} y_{1t} \\ y_{2t} \\ X_t \end{pmatrix} = A_0 + A_1 \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \\ X_{t-1} \end{pmatrix} + A_2 \begin{pmatrix} y_{1t-2} \\ y_{2t-2} \\ X_{t-2} \end{pmatrix} + A_3 \begin{pmatrix} y_{1t-3} \\ y_{2t-3} \\ X_{t-3} \end{pmatrix} + \begin{pmatrix} \mu_{1t} \\ \mu_{2t} \\ \mu_{3t} \end{pmatrix} \quad (4)$$

Where the endogenous variables y_{it} represent per capita CO₂ emissions, and a financial development indicator of interest. The exogenous variable X_t is GDP per capita.

Finally, for the purposes of determining the relative contribution of various financial development indicators and the GDP per capita variable to the evolution of carbon emissions, a Vector Error Correction Model (VECM) was also estimated and decomposition of variance was undertaken.

4. Results and discussion

The descriptive statistics of the entire data are presented in Table 2.

The graphical representation of CO₂ emissions with all the economic and financial development indicators is presented in Figs. 7–12.

The rapid increase in population, urbanization and industrial activity results in increased energy demand over the years. The actual impact of higher energy demand on emissions depends on both the amount of fuel used and efficiency of fuel usage. Pakistan has undergone quite rapid economic growth over the years and the per capita incomes rose substantially. Pakistan's emissions continued to grow at an increasing rate. A number of factors made the situation worse such as deforestation, lack of technological progress and a huge policy disconnect at the government level to take energy conservation initiatives. GDP per capita and CO₂ emissions showed strong upward trend and a notable degree of co-movement, showing that per capita income may be a major contributor in increasing CO₂ emissions in Pakistan.

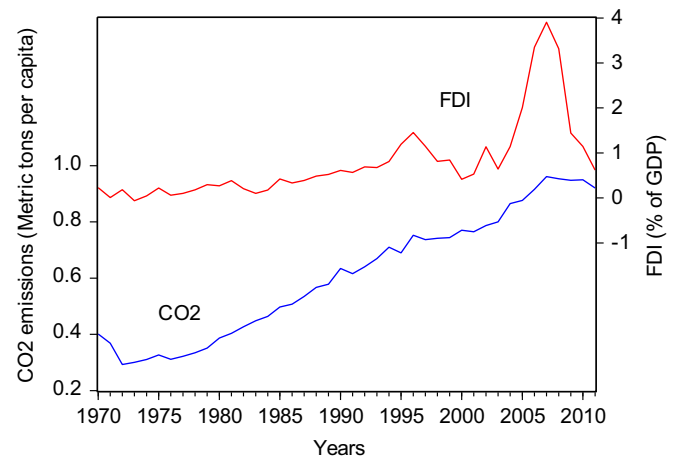


Fig. 7. Trends of variables in Pakistan_FDI (% of GDP).

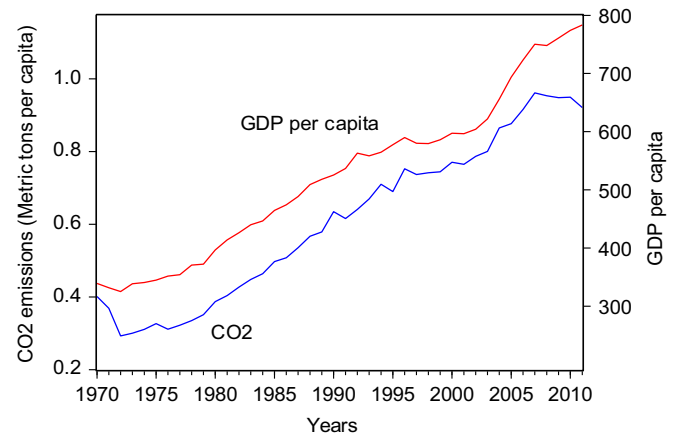


Fig. 8. Trends of variables in Pakistan_GDP per capita.

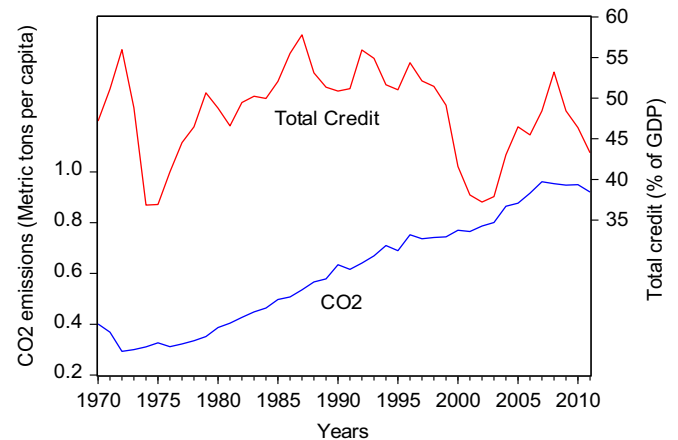


Fig. 9. Trends of variables in Pakistan_Total credit (% of GDP).

Table 2
Descriptive statistics.

Statistics	CO2 Emissions per capita	FDI (%)	Total credit (%)	Private sector credit (%)	Market capitalization (%)	Stock market turnover (%)	GDP Per capita
Mean	0.6142	0.7999	48.273	24.749	14.499	29.457	528.753
Maximum	0.9611	3.9044	57.786	29.841	49.076	128.646	783.304
Minimum	0.2929	-0.0632	36.851	18.371	2.3663	0.4601	325.259
St. Dev.	0.2207	0.9017	5.6439	2.8389	12.162	35.752	137.931

Note: All variables except per capita CO₂ emissions and GDP Per capita are expressed as percentages of GDP.

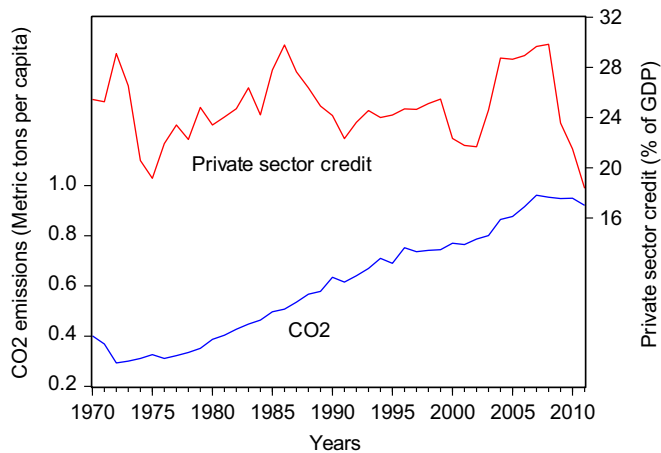


Fig. 10. Trends of variables in Pakistan_Private credit (% of GDP).

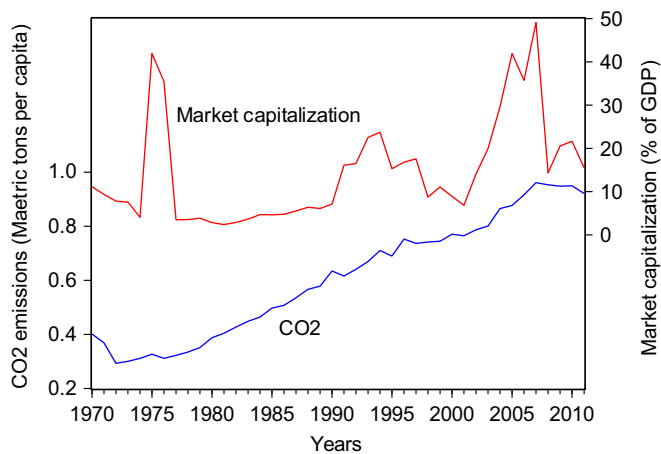


Fig. 11. Trends of variables in Pakistan_Market capitalization (% of GDP).

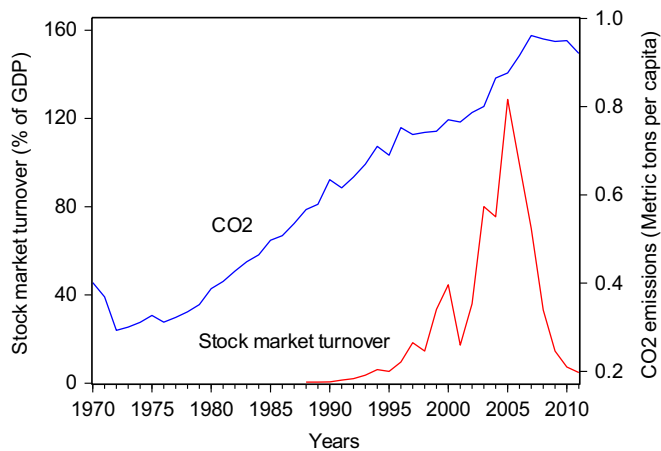


Fig. 12. Trends of variables in Pakistan_Stock market turnover (% of GDP).

4.1. Cointegration between CO₂ emissions and financial development indicators

The variables were first tested for cointegration by applying Autoregressive Distributed Lag (ARDL) bounds testing approach to determine cointegration developed by Pesaran et al. (2001). Then the short run dynamics were studied using the Error Correction Model (ECM).

Table 3

ARDL Cointegration Test Results (Full Sample: 1970–2011).

Model for estimation	F-Statistics (Restricted intercept and no trend)	Lags
CO ₂ emission per capita/ FDI percentage, Total credit percentage, Private sector credit percentage, GDP per capita, percentage of market capitalization	5.517***	4
Bounds Critical values	Significance Level	LCB UCB
	1%	3.881 5.241
	5%	2.780 3.989
	10%	2.323 3.376

Note: Based on Narayan (2004). All variables except per capita CO₂ emissions and GDP per capita are expressed as percentages of GDP.

Table 4

ARDL Cointegration Test Results (Reduced Sample: 1988–2011).

Model for estimation	F-Statistics (Restricted intercept and no trend)	Lags
CO ₂ emission per capita/ FDI percentage, Total credit percentage, Private sector credit percentage, GDP per capita, percentage of market capitalization, Stock turnover percentage.	7.209***	1
Bounds critical values	Significance Level	LCB UCB
	1%	3.976 5.691
	5%	2.794 4.148
	10%	2.334 3.515

Note: Based on Narayan (2004). All variables except per capita CO₂ emissions and GDP per capita are expressed as percentages of GDP.

Table 5

Long run model (Full sample).

Dependent variable = CO ₂ emissions per capita		
Variable	Coefficient	P-value
Constant	−0.2220	0.009
FDI (%)	−0.0025	0.851
Total credit (%)	−0.0019	0.144
Private sector credit (%)	0.0044	0.147
Market capitalization (%)	0.0001	0.823
GDP per capita	0.0016	0.000

Note: All variables except per capita CO₂ emissions and GDP Per capita are expressed as percentages of GDP.

Table 6

Long run model (Reduced sample).

Dependent variable = CO ₂ emissions per capita		
Variable	Coefficient	P-value
Constant	0.0729	0.626
FDI (%)	0.0082	0.576
Total credit (%)	−0.0038	0.088
Private sector credit (%)	0.0046	0.400
Market capitalization (%)	−0.0001	0.905
Stock Market Turnover (%)	0.0000	0.982
GDP per capita	0.0012	0.000

Note: All variables except per capita CO₂ emissions and GDP Per capita are expressed as percentages of GDP.

Table 7
ARDL Cointegration Test Results_Ln Model (Full Sample: 1970–2011).

Model for estimation	F-Statistics (Restricted intercept and no trend)	Lags
Ln CO ₂ emission per capita/ Ln (FDI percentage, Total credit percentage, Private sector credit percentage, GDP per capita, percentage of market capitalization)	6.451***	1
Bounds Critical values	Significance Level	LCB UCB
	1%	3.881 5.241
	5%	2.780 3.989
	10%	2.323 3.376

Note: Based on Narayan (2004). All variables except per capita CO₂ emissions and GDP per capita are expressed as percentages of GDP.

Table 8
ARDL Cointegration Test Results_Ln Model (Reduced Sample: 1988–2011).

Model For Estimation	F-Statistics (Restricted intercept and no trend)	Lags
Ln CO ₂ emission per capita/ Ln (FDI percentage, Total credit percentage, Private sector credit percentage, GDP per capita, percentage of market capitalization, Stock turnover percentage.)	6.609***	2
Bounds Critical values	Significance Level	LCB UCB
	1%	3.976 5.691
	5%	2.794 4.148
	10%	2.334 3.515

Note: Based on Narayan (2004). All variables except per capita CO₂ emissions and GDP per capita are expressed as percentages of GDP.

ARDL results for full and reduced samples are presented in Tables 3 and 4 respectively. The results for both samples revealed that the calculated F-statistics exceeded the critical values at 1% level of significance. We concluded that the per capita CO₂ emissions was cointegrated with all the variables at 1% significance level.

After the evidence of cointegration between variables, the long run relationship was estimated for both samples. There was a positive significant relationship between GDP per capita and CO₂ emissions at 1% in full sample (See Table 5). None of the financial development indicators showed any significant long run relationship with per capita CO₂ emissions in full sample model. In the full sample, therefore, we did not find evidence of emission mitigating role of financial development.

We re-estimated the long run relationship between per capita CO₂ emissions and economic and financial variables for reduced sample, and the results are reported in Table 6. The findings revealed that the impact of percentage of total credit in GDP on per capita CO₂ emissions is significant and negative at 10% level. The increase in total credit was associated with more investment in energy efficient, cleaner technologies that resulted in improved environmental performance. These results are in line with Tadesse (2005), Tamazian et al. (2009), Tamazian and Bhaskara Rao (2010) and Jalil and Feridun (2011).

The relationship between GDP per capita and CO₂ emissions was positive and significant at 1% level in both samples. Thus as per capita income rose, there was increased demand for energy, which mainly came from fossil fuels usage that contributed to CO₂ emissions. The findings are similar to that of Ozturk and Acaravci (2010) for Turkey, Saboori and Soleymani (2011) for Iran, Shahbaz et al. (2011) for Pakistan and Shahbaz et al. (2011) for South Africa.

The differences in results between the full and the reduced

Table 9
Long run_Ln model (Full sample).

Dependent Variable = Ln CO ₂ emissions per capita		
Variable	Coefficient	P-value
Constant	−9.613	0.000
Ln FDI (%)	−0.009	0.653
Ln Total credit (%)	−0.037	0.747
Ln Private sector credit (%)	0.152	0.227
Ln Market capitalization (%)	0.021	0.219
Ln GDP per capita	1.389	0.000

Note: All variables except per capita CO₂ emissions and GDP Per capita are expressed as percentages of GDP.

Table 10
Long run_Ln model (Reduced sample).

Dependent Variable = Ln CO ₂ emissions per capita		
Variable	Coefficient	P-value
Constant	−5.856	0.000
Ln FDI (%)	0.022	0.278
Ln Total credit (%)	0.012	0.874
Ln Private sector credit (%)	−0.092	0.256
Ln Market capitalization (%)	−0.020	0.166
Ln Stock Market Turnover (%)	0.032	0.000
Ln GDP per capita	0.905	0.000

Note: All variables except per capita CO₂ emissions and GDP Per capita are expressed as percentages of GDP.

sample in the above analysis, led us to consider alternative model specifications.³ The model with natural log of all the variables gave the most statistically improved results. Tables 7 and 8 present the ARDL cointegration results of this model in both full and reduced sample showing that the models are cointegrated at 1% significance level. After the evidence of cointegration, long run models are estimated for both samples.

In case of full sample, the relationship between CO₂ emissions and GDP per capita was positive and significant at 1% level (See Table 9). These findings are similar to the previously tested long run model for full sample, in that none of the financial development indicators showed any significant long run relationship with CO₂ emissions. Thus we conclude that the financial development did not seem to have been the key drivers of CO₂ emissions over the full sample period. The reasons for this rather peculiar finding seem to lie in the low levels of financial development in Pakistan during the early period of the financial development. As a corollary, we also concluded that it is not only the differences in the level of financial development among countries, but also the evolution of financial development over time within the same country, that determines the long-term behavior of carbon emissions. In particular, at low levels of financial development, the rise in income per capita appears to be the main driver of emissions, and neither the scale nor the efficiency of financial intermediation matters. These findings are in sharp contrast to those reported in Zhang (2011) who found that both the scale and efficiency of financial intermediation impact evolution of emissions. Thus the results of our study serve to underscore the need for a more cautious approach to drawing inference about the relationship between emissions and financial development, and not extrapolating the findings from the larger economies with more developed and structurally different financial sectors to smaller economies where the development of the financial sectors may be

³ We are grateful to an anonymous referee for raising this issue.

Table 11
Error Correction Model (Full Sample).

Dependent Variable = ΔCO_2		
Variable	Coefficient	P-value
Constant	0.0039	0.471
ΔFDI (% of GDP)	0.0088	0.324
$\Delta \text{Total credit}$ (% of GDP)	−0.0019	0.183
$\Delta \text{Private sector credit}$ (% of GDP)	0.0030	0.242
$\Delta \text{Market capitalization}$ (% of GDP)	0.0003	0.347
$\Delta \text{GDP per capita}$	0.0009	0.025
ECT (−1)	−0.2981	0.017
R-Square	0.4319	
Adjusted R-Square	0.3316	
S.E. of regression	0.0219	
AIC	−4.649	
SC	−4.357	
Durbin-Watson	2.0235	
Diagnostic and Stability tests:		
	Coefficient	P-value
Jarque-Bera Normality Test	3.769	0.152
Lagrange Multiplier (LM) Test	1.016	0.373
Heteroskedasticity Test	0.583	0.741
CUSUM	Stable (5%)	
CUSUM of square	Stable (5%)	

Table 12
Error Correction Model (Reduced Sample).

Dependent Variable = ΔCO_2		
Variable	Coefficient	P-value
Constant	0.0118	0.149
ΔFDI (% of GDP)	0.0014	0.892
$\Delta \text{Total credit}$ (% of GDP)	−0.0027	0.277
$\Delta \text{Private sector credit}$ (% of GDP)	0.0075	0.066
$\Delta \text{Market capitalization}$ (% of GDP)	0.0006	0.379
$\Delta \text{Stock Market turnover}$ (% of GDP)	−0.0002	0.578
$\Delta \text{GDP per capita}$	0.0005	0.392
ECT (−1)	−0.5491	0.025
R-Square	0.5728	
Adjusted R-Square	0.3733	
S.E. of regression	0.0214	
AIC	−4.5820	
SC	−4.1870	
Durbin-Watson	2.6870	
Diagnostic and stability tests:		
	Coefficient	P-value
Jarque-Bera Normality Test	0.695	0.706
Lagrange Multiplier (LM) Test	5.471	0.019
Heteroskedasticity Test	0.291	0.947
CUSUM	Stable (5%)	
CUSUM of square	Stable (5%)	

at early stages.

The long run relationship for reduced sample was estimated and the results are reported in Table 10. Both GDP per capita and stock market turnover as percentage of GDP are significant and positive at 1% level. We considered this to be a relatively more robust model as compared to the previous long run model for reduced sample because the results were statistically better determined. This model suggested that both GDP per capita and stocks traded were the main drivers of CO_2 emissions in Pakistan. The findings are consistent with Demircuc-Kunt et al. (2012) who suggest that as the countries develop, financial sector as a whole becomes more important, and within the financial sector, the role of financial markets becomes more important relative to the role of banks. The significance of stocks traded only in the reduced sample pointed towards the same conclusion as mentioned earlier, that the later period of the sample was characterized by more financial development especially in stock markets [also see Figs. 4 and 5]. For comparing the results of the two models, estimated respectively on the full and the reduced samples, we note that the stock traded variable was not included in the model estimated on full sample because sufficiently long time series for this variable was not available. This could have led to omitted variable bias. We conclude, therefore, that the reduced sample models were better specified compared to their full sample counterparts.

Next we consider the short run dynamics of the two models. Short run results of ECM for full and reduced samples are reported in Tables 11 and 12 respectively. The lag length selected through SC, was zero.

The full sample results indicated that in the short run, the changes in GDP per capita had significant positive impact on the changes in per capita CO_2 emissions. These results are consistent with Ozturk and Acaravci (2010) for Turkey, Jalil and Faridun (2011) for China, Shahbaz et al. (2011) for Pakistan, Saboori and Suleymani (2011) for Iran, Shahbaz et al. (2011) for South Africa. The lagged ECM term was found to be negative and significant at 5% level. The value of lagged ECM term was −0.2981. This suggested that 29.81% of the change in per capita CO_2 emissions were corrected each year.

The reduced sample ECM shows that in the short run, the changes in the share of private sector credit in GDP had significant positive impact on the changes in per capita CO_2 emissions. These results suggested that an increase in the private sector credit was associated with more energy consumption and emissions that leads to poor environmental performance, in the short run. Interestingly, the change in GDP per capita was found not to be statistically significant in explaining the changes in carbon emissions. The sign of the lagged ECM term was negative and significant at 5% level. The value of lagged ECM term was −0.5491 which indicated that 54.91% of the change in per capita CO_2 emissions was corrected each year.

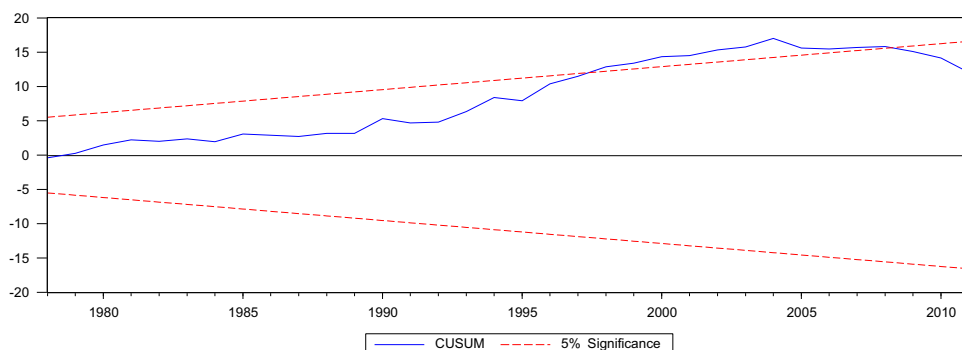


Fig. 13. Plot of Cumulative Sum of Recursive Residuals (Full Sample).

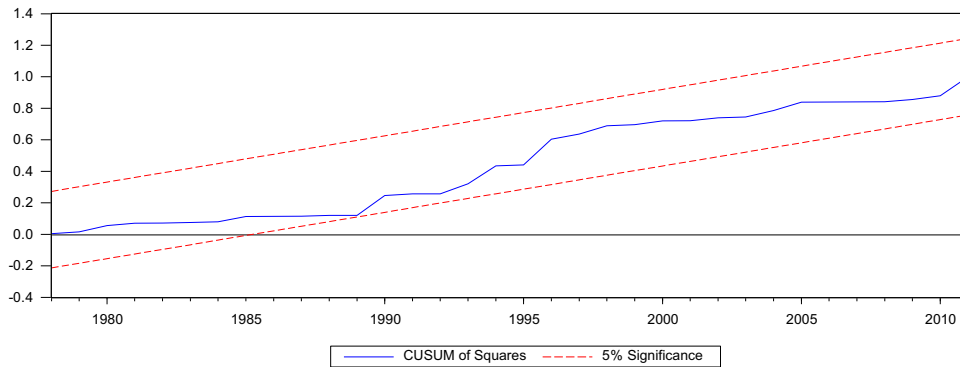


Fig. 14. Plot of Cumulative Sum of Squares of Recursive Residuals (Full Sample).

4.1.1. Diagnostic and stability tests

To ensure the goodness of fit of ECM, stability tests such as cumulative sum of recursive residuals (CUSUM), cumulative sum of squares of recursive residuals (CUSUMsq) and diagnostic tests such as Jarque-Bera test, Langrage Multiplier (LM) and White Heteroskedasticity test were conducted.

The Cumulative Sum (CUSUM) and Cumulative Sum of Squares (CUSUMSQ) tests were conducted to investigate the stability of long run and short run parameters. Graphically, the two test statistics are plotted within the two bounds of 5% significance level (see Figs. 13 and 14). If the plots lie between these 5% levels, the null hypothesis of stable parameters is accepted otherwise if the plot significantly crosses the 5% level, the null is rejected. The CUSUM and CUSUMSQ plots for full sample show that both the models are generally stable.

Figs. 15 and 16 provide the plots for CUSUM and CUSUMSQ respectively for reduced sample. In reduced sample as well, model stability seemed to hold.

In diagnostic tests, the model is tested for normality of residual terms, serial correlation, and Heteroskedasticity. Jarque-Bera test was used to test for the normality of residuals, Langrage Multiplier (LM) test was used to check the serial correlation among the residual terms and White Heteroskedasticity test was employed to determine whether variance of the residual terms changed over time.

The full sample results suggested that the model passes all the diagnostic tests i.e. the residual terms were normally distributed, there was no serial correlation among them and the functional form of model was also well specified. Similarly for the reduced sample, the residual terms were found to be normally distributed, functional form of model was also well specified and the null hypothesis of no serial correlation was rejected in this case.

4.2. Causality between CO₂ emissions and financial development indicators

The augmented VAR (Toda and Yamamoto, 1995) in levels of variables was implemented to test for Granger Causality.

Augmented Dickey-Fuller test (ADF) was used to identify the order of integration in present study. Schwarz information criterion (SC) was used to select the optimal lag length for the model. SC was preferred to select the optimum lag length as compared to other information criteria because it selects the most parsimonious model in case of finite small samples.

The results of these tests for the full sample are presented in Table 13.

The unit root tests performed on variables in levels, suggested that the private sector credit (% of GDP) and FDI (% of GDP) were stationary series. The variables representing, respectively, the shares in GDP of the total credit, stock market capitalization, stock market turnover as well as CO₂ emissions and GDP per capita, all contained unit roots.

All the non-stationary variables were found to become stationary after first differencing so these variables were integrated of order one i.e. $I(1)$. Table 14 shows the first differencing results of the variables.

The augmented VAR framework is used for pair-wise testing of only two endogenous variables-CO₂ emissions per capita and one financial sector indicator-while controlling for the effect of an exogenous variable (GDP per capita) in the relationship. The Granger causality test results for $(k + d_{\max})$ th order of VAR (Full sample) are shown in Table 15 where k is the original order of VAR.

The results showed that FDI as percentage of GDP Granger caused per capita CO₂ emissions at 10% significance level but the reverse was not true. We failed to reject all other null hypothesis of Granger non-causality.

The Granger causality test results for reduced sample are shown in Table 16.

In reduced sample case, Private sector credit (% of GDP) Granger caused per capita CO₂ emissions at 1% significance level. Moreover, stock market turnover (% of GDP) Granger caused CO₂ emissions at 5% level. The impact of financial development indicators improved in the latter years (reduced sample). As the level of financial development increased over time, its impact on CO₂ emissions

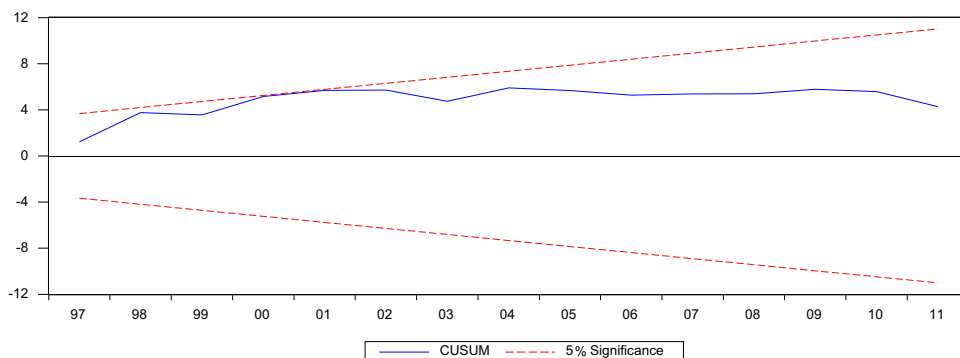


Fig. 15. Plot of Cumulative Sum of Recursive Residuals (Reduced Sample).

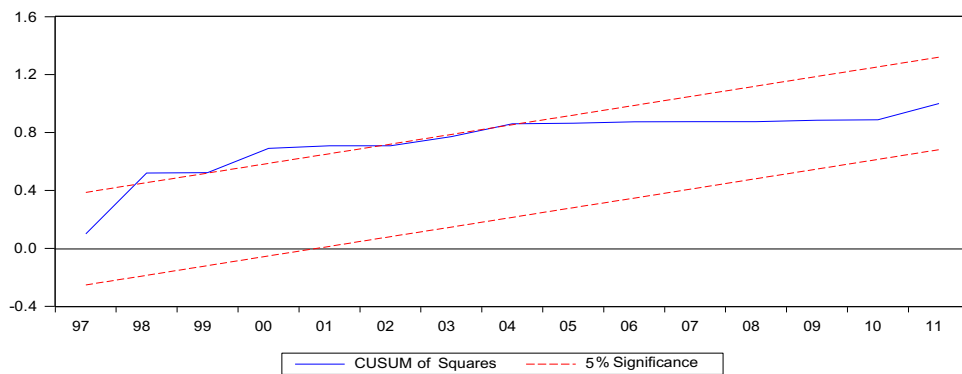


Fig. 16. Plot of Cumulative Sum of Squares of Recursive Residuals (Reduced Sample).

Table 13

Unit root tests in levels of variables.

ADF Test at Level with Trend and Intercept		
Variables	T-Statistics	P-values*
CO ₂ Emissions	−2.217	0.467
Total credit (% of GDP)	−2.280	0.434
Credit to private sector (% of GDP)	−3.445	0.060
FDI (% of GDP)	−4.228	0.009
Market capitalization (% of GDP)	−2.783	0.211
Stock market turnover (% of GDP)	−1.978	0.579
GDP per capita	−2.137	0.509
ADF Test at Level with Intercept only		
Variables	T-Statistics	P-values*
CO ₂ Emissions	−0.090	0.777
Total credit (% of GDP)	−2.320	0.171
Credit to private sector (% of GDP)	–	–
FDI (% of GDP)	–	–
Market capitalization (% of GDP)	−2.378	0.154
Stock market turnover (% of GDP)	−1.946	0.306
GDP per capita	2.107	0.969
ADF Test at Level with no Intercept and no Trend		
Variables	T-Statistics	P-values*
CO ₂ Emissions	2.598	0.997
Total credit (% of GDP)	−0.698	0.408
Credit to private sector (% of GDP)	–	–
FDI (% of GDP)	–	–
Market capitalization (% of GDP)	−1.156	0.222
Stock market turnover (% of GDP)	−1.315	0.168
GDP per capita	2.848	0.999

* MacKinnon (1996) one-sided p-values.

Table 14

ADF Test at First difference.

Model with no Intercept and no Trend		
Variables	T-statistics	P-values
CO ₂ Emissions	−2.067	0.039
Total Credit (% of GDP)	−3.647	0.001
Credit to private sector (% of GDP)	–	–
FDI (% of GDP)	–	–
Market capitalization (% of GDP)	−4.235	0.000
Stock market turnover (% of GDP)	−2.112	0.036
GDP per capita	−1.765	0.074

became more evident.

4.3. Variance decomposition based on VECM

Finally, we explored the extent of contribution of different economic and financial variables to the changes in CO₂ emissions. This was accomplished by performing a variance decomposition

analysis based on VECM. The lag length was selected through SC, the estimated trivariate VECM is as follows;

$$\Delta Y_{1t} = \alpha_1 + \alpha_2 ECT_{t-1} + \sum_{k=0}^n \alpha_{3k} \Delta Y_{2t-k} + \sum_{k=0}^n \alpha_{4k} \Delta Y_{3t-k} + \mu_{1t} \quad (5)$$

$$\Delta Y_{2t} = \alpha_5 + \alpha_6 ECT_{t-1} + \sum_{k=0}^n \alpha_{7k} \Delta Y_{1t-k} + \sum_{k=0}^n \alpha_{8k} \Delta Y_{3t-k} + \mu_{2t} \quad (6)$$

$$\Delta Y_{3t} = \alpha_9 + \alpha_{10} ECT_{t-1} + \sum_{k=0}^n \alpha_{11k} \Delta Y_{1t-k} + \sum_{k=0}^n \alpha_{12k} \Delta Y_{2t-k} + \mu_{3t} \quad (7)$$

Where 'k' denotes the lag length, the error correction term (ECT) indicates the speed of adjustment of the variable after its deviation from the long run equilibrium, the short run dynamics of the model are represented by coefficients of the lagged difference terms.

For the full sample, we found evidence of only one variable i.e. GDP per capita having a long run equilibrium relationship with CO₂ emissions. However, for the reduced sample period that represented relatively higher level of financial sector development, the stock turnover share in GDP and GDP per capita both had long run equilibrium relationship with carbon emissions. A VECM of these indicators was formulated and variance decomposition was performed to understand their respective contribution in evolution of per capita CO₂ emissions. The results are shown in Fig. 17.

The results indicated that GDP per capita had large influence to the extent of 13.19% of the total variance of per capita carbon emissions, while the influence of stock turnover as percentage of GDP was only 1.57%. We concluded that financial development did not contribute much in explaining the variance of per capita carbon emissions. This may be because financial development is in its initial stages in the country. On the other hand, the economic growth contributes largely towards the changes in per capita carbon emissions, indicating that extensive output growth leads to more fossil fuel consumption that contributes towards higher level of emissions.

5. Conclusion and policy recommendations

The study found that per capita CO₂ emissions were cointegrated with financial development indicators and per capita GDP. The estimated long run model for full sample showed that only GDP per capita had statistically significant impact on CO₂ emissions, causing emissions to increase with the standards of living. The more robust estimated long run model (see Table 10) for the reduced sample showed that both GDP per capita and percentage of stock turnover in GDP had statistically significant impact on CO₂ emissions. The per capita CO₂ emissions increased

Table 15
Granger Causality Results (Full sample).

Null hypothesis	Lag length 'k'	k+ d _{max}	Chi-square statistic	Prob.
CO ₂ does not G.C the change of FDI	3	4	3.081	0.379
FDI does not G.C the change of CO ₂			6.772*	0.079
CO ₂ does not G.C the change of Total Credit	1	2	0.349	0.554
Total credit does not G.C the change of CO ₂			0.393	0.530
CO ₂ does not G.C the change of Private sector credit	1	2	1.064	0.302
Private sector credit does not G.C the change of CO ₂			0.882	0.348
CO ₂ does not G.C the change of Market capitalization	1	2	0.319	0.572
Market capitalization does not G.C the change of CO ₂			0.208	0.648

as the share of stock turnover in GDP and GDP per capita rose. The reduced sample corresponds to a period of relatively higher degree of financial sector development in Pakistan; it is noteworthy that it was during this period that we found that a financial development indicator was statistically significant in explaining the evolution of carbon emissions.

We interpreted the long run results to suggest that economic growth increased the energy consumption and hence caused more emissions. Moreover, the development in financial sector also contributed towards increase in emissions. The development of stock market eased the liquidity constraints faced by the listed firms, allowing them to expand output, increasing energy consumption and hence CO₂ emissions.

The short run dynamics showed that only the growth in per capita incomes was associated with rise in the per capita CO₂ emissions, in the full sample, while none of the other variables had significant relationship with them. In the case of the reduced sample, the percentage of private sector credit had significant relationship with CO₂ emissions i.e. CO₂ emissions increased with the increase of private sector loans. Thus in full sample only GDP per capita was significant in both the long run and the short run ECM. While in reduced sample, in addition to GDP per capita, the carbon emissions were statistically significantly impacted in the long run by stock market turnover as percentage of GDP and by private sector credit in the short run ECM.

The Granger causality showed that holding per capita GDP as exogenous in the full sample, only the share of FDI in GDP Granger caused per capita carbon emissions at 10% level. This implies that in an emerging economy like Pakistan, FDI had a unidirectional causal relationship with emissions. No other indicator was found to Granger cause carbon emissions per capita. In the reduced sample, the impact of financial sector development became evident. In this case, the percentage of private sector credit and the stock market turnover both Granger caused per capita carbon emissions. These results suggest that in the emerging economy considered in the paper, the financial intermediation efficiency and stock market efficiency Granger caused per capita CO₂ emissions. These results were obtained only in the reduced sample period that corresponded to a relatively higher degree of financial

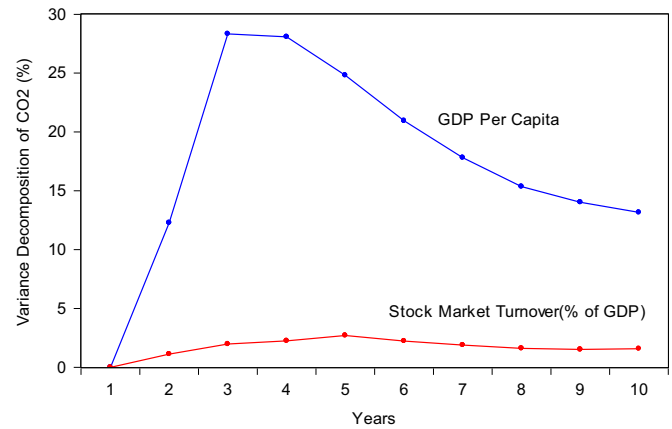


Fig. 17. Variance decomposition results.

development. This suggests that among the emerging economies group in general, the role that the financial sector could play in bringing about desirable environmental outcomes depends on the maturity of the financial sector itself.

The variance decomposition results revealed that the growth in per capita GDP explained more substantial amount of variance of per capita CO₂ emissions (13.19%) compared to that which was explained by share of stock traded to GDP (1.57%).

Several policy recommendations follow from the above analysis. First, the CO₂ emissions rise as per capita income rises, increasing the carbon footprint of the economy. Therefore, rising per capita incomes need to be factored into sustainable development plans. Second, in the emerging economy in question, and perhaps in many other emerging economies where the structural transformation of the financial sector is in its early stages, financial development does not aid in mitigating of emissions, rather it increases it. This is in sharp contrast to larger more financially developed emerging economies as reported by earlier studies such as Zhang (2011). So the respective national governments need to play more active roles for effective mitigation. In particular, the governments in these economies should devise comprehensive

Table 16
Granger Causality Results (Reduced sample).

Null hypothesis	Lag length 'k'	k+ d _{max}	Chi-square statistic	Prob.
CO ₂ does not G.C the change of FDI	3	4	2.386	0.496
FDI does not G.C the change of CO ₂			1.705	0.635
CO ₂ does not G.C the change of Total Credit	1	2	0.244	0.621
Total credit does not G.C the change of CO ₂			0.076	0.782
CO ₂ does not G.C the change of Private sector credit	1	2	0.028	0.865
Private sector credit does not G.C the change of CO ₂			7.287***	0.007
CO ₂ does not G.C the change of Market capitalization	1	2	0.423	0.515
Market capitalization does not G.C the change of CO ₂			1.914	0.166
CO ₂ does not G.C the change of Stock turnover	1	2	0.882	0.347
Stock turnover does not G.C the change of CO ₂			5.524**	0.019

and realistic mitigation strategies that blend economic incentives with regulatory measures aimed at mitigation.

This requires several mitigation-enabling activities. The foremost among these is building capacities of national institutions for reliable information collection and analysis so as to be able to estimate baseline emissions, and to make emissions and cost forecasts under alternative mitigation scenarios. Moreover, the countries should identify important data gaps that need to be filled for identifying viable mitigation options, and make efforts to fill them. In the case of Pakistan, these gaps include information on energy end-uses, especially for transportation and domestic use sector.⁴

The preparation and successful implementation of mitigation strategies requires coordination among several stakeholders such as various government ministries, research institutes, universities and industry. These pre-requisites, and the necessary political will, did not exist in the past [Khan and Baig (2009)]. Pakistan missed the target date for submission of the Intended Nationally Determined Contributions (INDCs) for emissions reduction before COP-21 conference. In view of this, greater cooperation is needed among all the key players.

Pakistan's energy and transport sectors are the largest source of emissions, along with agriculture and industrial sector [Khan and Baig (2009)]. Government should emphasize on introducing efficient technologies in these sectors to reduce the cost and fuel usage and increase productivity thus accelerating economic growth and yielding environmental benefits. Other actions could be mass tree plantation campaigns across country, energy generation from renewable sources including hydro, solar and wind, carbon taxes, subsidies for adoption of green technologies, tradable carbon quotas, legislated rules and regulations such as minimum fuel efficiency standards for vehicles, and energy efficiency provisions incorporated into housing regulations.

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⁴ For example, data on vehicle kilometers traveled by vehicle type is from studies that are 10 years old, e.g. ENERCON (2003), and JICA (2006). Updated information is not available.